

AI GOVERNANCE CASE STUDY

Autonomous Background Agents and Structural Failures in AI Governance

Case subject: KAIROS

Source: Claude Code source artifacts (npm package, March 31, 2026)

King Che Magnusson

OWASP HCI Cognitive Layer Project

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Epistemic note

This analysis is based on publicly available source code artifacts and reported findings following the March 31, 2026 exposure of Claude Code internals.

Certain interpretations of system behaviour are inferred from code structure and internal comments and should be understood as analytical reconstruction rather than confirmed deployed functionality.

1. System description

On March 31, 2026, Anthropic published version 2.1.88 of Claude Code on npm. A large source map file (approx. 59.8 MB) was unintentionally included, exposing a substantial portion of the internal TypeScript codebase.

Within this material, a system referred to as **KAIROS** appeared across numerous references. The system is feature-flagged and not active in public builds, but represents a significant architectural shift from reactive tooling to a persistent, proactive agent model.

System classification

- **Type:** Autonomous agent system with persistent execution
- **Interaction mode:** Background, non-user-triggered
- **Memory model:** Persistent, self-modifying
- **Autonomy:** Context-adaptive
- **Operational domain:** Software development lifecycle (repositories, CI/CD environments)

Core characteristics

- **Persistent execution:** Runs as a background daemon independent of active user sessions
- **Tick mechanism:** Periodic activation cycles triggering decisions (act vs idle)
- **AutoDream:** Memory consolidation process transforming observations into persistent knowledge
- **Adaptive autonomy:** Increased independence when user is inactive

- **Repository monitoring:** Reacts to repository events via webhooks
- **Undercover mode:** Suppresses disclosure of AI origin in outputs
- **Memory model:** Append-only logs during runtime, periodically consolidated

2. Risk mapping

2.1 Transparency failure

Mechanism: Intentional suppression of system identity in output layer

KAIROS appears to include functionality designed to obscure its AI origin in outputs (e.g. commits, reviews). This is not incidental but embedded in system behaviour.

Implications:

- Users may be unable to distinguish human-generated from AI-generated outputs
- Organisations may deploy autonomous systems without full awareness
- Audit trails lose integrity if origin attribution is obscured
- Memory consolidation may alter historical traceability

2.2 Accountability diffusion

Mechanism: Asynchronous autonomous execution without explicit human trigger

KAIROS can act independently, without direct user initiation, particularly during periods of user inactivity.

Resulting condition:

A breakdown in attribution of responsibility:

- No clear human decision point
- No explicit approval chain
- No stable linkage between action and accountable actor

This creates a structural accountability gap rather than an operational error.

2.3 Cognitive offloading and decision drift

Mechanism: Gradual establishment of system reliability expectations (anchoring), reducing likelihood of human intervention (automation bias)

Over time, consistent autonomous performance may lead users to:

- Trust outputs without verification
- Lose situational awareness
- Delegate implicit decision authority to the system

This is not isolated automation bias but a **system-level shift in cognitive control distribution**.

3. Legal mapping

3.1 EU AI Act

Article	Requirement	Observed implication
Art. 9	Risk management	Lack of visible lifecycle risk controls for autonomous operation
Art. 13	Transparency	Undercover behaviour appears incompatible with disclosure requirements if deployed without user awareness
Art. 14	Human oversight	Adaptive autonomy reduces oversight precisely when users are absent

3.2 GDPR

Article	Requirement	Observed implication
Art. 5	Transparency, purpose limitation	Persistent profiling may expand beyond initial purpose
Art. 6	Legal basis	Behavioural profiling requires explicit justification
Art. 22	Automated decisions	Autonomous actions may impact third parties without awareness

4. Failure modes

4.1 Temporal failure: Night-time decision drift

System autonomy increases during user absence.

Scenario:

System behaviour evolves through memory consolidation without human validation, leading to altered system reasoning over time.

4.2 Attribution failure: Hidden authorship

AI-generated outputs are indistinguishable from human outputs.

Implications:

- Auditability breaks
- Human review requirements may be bypassed in practice
- Organisations lose visibility into actual AI usage

4.3 Epistemic failure: Memory mutation

Mechanism: Transformation of uncertain observations into stable internal “facts”

Implication:

- Historical reasoning cannot be reconstructed
- Decision traceability is degraded

- System knowledge evolves without audit visibility

5. Controls

Standard AI controls are insufficient for persistent autonomous agents. The following controls address identified failure modes directly.

5.1 Mandatory attribution layer

Description: Cryptographic signing of all AI actions

Purpose:

- Ensure traceability of AI-generated outputs
- Prevent misattribution

Addresses:

- Art. 13 (transparency)
- Auditability requirements
- Hidden authorship

5.2 Mandatory review gates

Description: No autonomous irreversible actions without explicit human approval

Implementation:

- All commits require authenticated human approval
- Destructive actions blocked
- Autonomy restricted during user absence

Addresses:

- Art. 14 (human oversight)
- Night-time decision drift
- Accountability gaps

5.3 Immutable memory logs

Description: Separation of working memory and immutable audit logs

Implementation:

- Append-only audit logs
- Memory consolidation cannot overwrite logs
- Full traceability of reasoning changes

Addresses:

- Epistemic failure

- Audit reconstruction
- Compliance verification

5.4 Autonomy caps

Description: Static, policy-defined autonomy levels

Implementation:

- No dynamic escalation based on context
- Explicit configuration required
- Human approval for escalation

Addresses:

- Accountability diffusion
- Oversight degradation
- Cognitive offloading risks

Control mapping

Failure mode	Primary control	Secondary control
Temporal drift	Autonomy caps	Review gates
Hidden authorship	Attribution layer	Immutable logs
Memory mutation	Immutable logs	Autonomy caps
Accountability gap	Review gates	Attribution layer
Cognitive offloading	Autonomy caps	Review gates

6. Summary assessment

KAIROS represents a class of systems where:

- autonomy is persistent
- decision-making is asynchronous
- system identity may be obscured
- memory evolves without full traceability

These characteristics introduce **structural failure modes** in:

- transparency
- accountability
- human oversight

The significance of this case is not tied to intent or specific deployment, but to architectural patterns that can emerge in advanced AI systems.

Conclusion

Effective governance of autonomous agents requires:

- **technical enforcement**, not policy reliance
- **traceability by design**, not post-hoc reconstruction
- **explicit control boundaries**, not adaptive autonomy

Controls such as attribution layers, review gates, immutable logs, and autonomy caps should be considered **minimum architectural requirements**, not optional safeguards.